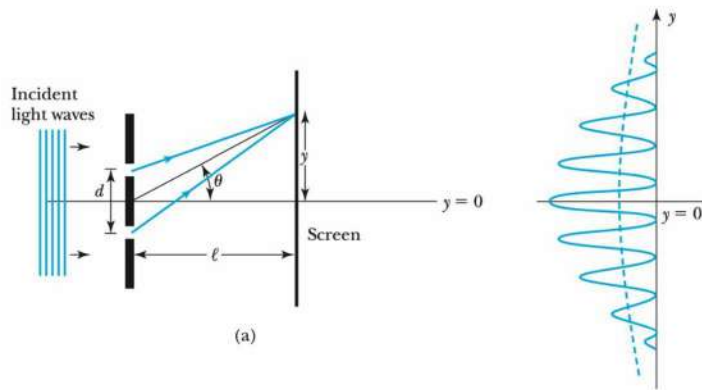


Particle-wave duality

Interference, double-slit experiment

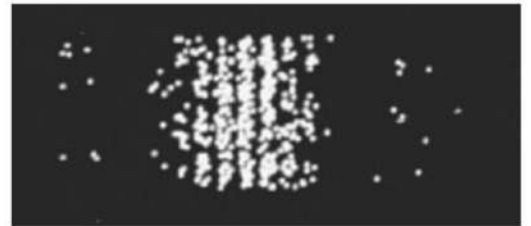


small intensity



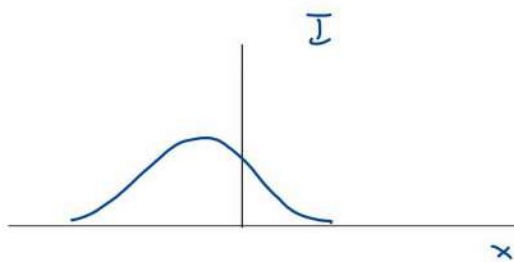
(b) 100 counts

large intensity

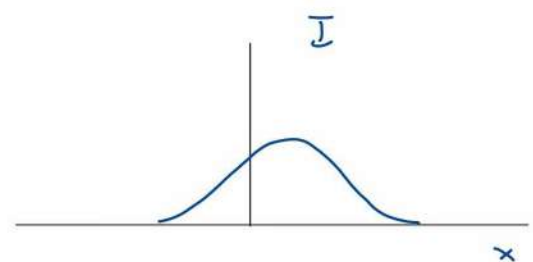


(c) 500 counts

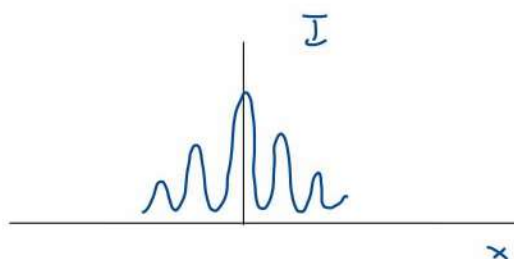
only left slit open



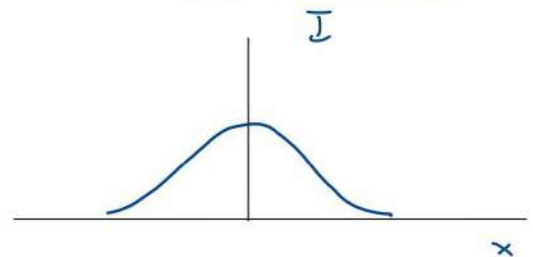
only right slit open



both slits open



both slits open
but observed



position and momentum: statistical variables
complementary: cannot both be determined
simultaneously with infinite
precision

(Cloud chamber: determines both, but not accurately)

measurement of $x \rightarrow$ random response
 $P(x)$

$$\langle x \rangle = \int dx P(x)$$

$$\langle x^2 \rangle = \int dx P(x) x^2 \Rightarrow \Delta x^2 = \langle x^2 \rangle - \langle x \rangle^2$$

Heisenberg:

$$\Delta p \cdot \Delta x \geq \frac{\hbar}{2}$$

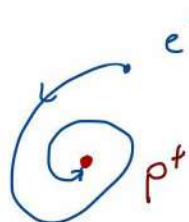
uncertainty principle
(rigorous mathematical statement)

time - energy ? $\downarrow$

No such rigorous relation!

important consequence $\rightarrow$ atoms do not
collapse

Classically



Q.M.

suppose e^- is confined to
 $r \lesssim a$

$$\Rightarrow \Delta p \gtrsim \frac{\hbar}{a} \Rightarrow K_{\min} \gtrsim \frac{\hbar^2}{2m \cdot a^2}$$

total energy

$$\langle E \rangle = \left\langle \frac{p^2}{2m} - \frac{d}{r} \right\rangle \sim \frac{\hbar^2}{2m \cdot a^2} - \frac{d}{a}$$

minimum energy:

$$\frac{\partial E(a)}{\partial a} = -\frac{\hbar^2}{m \cdot a^3} + \frac{2}{a^2} = 0$$

$$\Rightarrow a_0 = \frac{\hbar^2}{m \cdot 2} = \frac{\hbar^2 \cdot 4\pi\epsilon_0}{m \cdot e^2} = a_B \quad !$$

Bohr radius !

Wave function

- electron propagates as wave $\psi(x, t)$

free electron: $\psi(x, t) = \int A(k) e^{i k \cdot x} e^{-i \frac{\hbar^2 k^2}{2m} \cdot t} dk$

$\vec{E} \leftrightarrow \psi$ $\uparrow$ complex

- Intensity of light $\sim \vec{E}^2(x, t)$

$\Rightarrow$ "intensity" of $e^- \rightarrow |\psi|^2(x, t)$

$\uparrow$
probability !

$$P(x, t) = |\psi(x, t)|^2$$

$P(x, t) \cdot dx$: probability of finding particle
in $[x, x+dx]$

$$\int dx P(x, t) \equiv 1 \quad \Leftrightarrow \quad \int dx |\psi(x, t)|^2 = 1$$

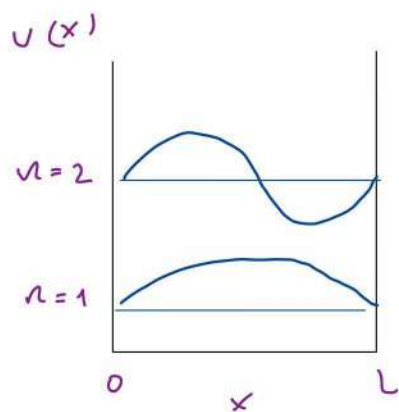
wave function
normalization

- $\psi(x, t)$ satisfies a wave equation (Schrödinger)

Nature is intrinsically probabilistic ...

Measurement ?

particle in box:



standing waves $\leftrightarrow$ bound states

$$\psi(x, t) = \varphi(x) \cdot \phi(t)$$

must vanish at $x = 0, L$

$$\Rightarrow \varphi(x) \propto \sin\left(\frac{x}{L} \cdot \pi \cdot n\right) = \sin\left(2\pi \cdot \frac{x}{\lambda}\right)$$

$$\Rightarrow \lambda = \frac{2 \cdot L}{n} \quad \Rightarrow p = \frac{h}{\lambda} = n \frac{h}{2 \cdot L}$$

$$\Rightarrow E = \frac{p^2}{2m} = n^2 \frac{h^2}{8 \cdot m \cdot L^2} \quad n = 1, 2, 3, \dots$$

discrete energies!

for standing wave $P(x) = |\varphi(x)|^2 \cdot |\phi(t)|^2$
must be constant $\equiv 1$

$$\text{then} \quad \int dx P(x) = \int dx |\varphi(x)|^2 = 1$$

$$\Rightarrow \varphi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi \cdot n}{L} \cdot x\right)$$

$$E_n = \frac{h^2}{8 \cdot m \cdot L^2} \cdot n^2$$

Schrodinger's equation

Wave equation

Requirement:

1. de Broglie and Einstein relations
 $p = \frac{h}{\lambda}$ $E = h \cdot f = \hbar \cdot \omega$
2. group velocity $\leftrightarrow$ classical velocity
3. interference $\sim$ homogeneous linear

ψ : amplitude of matter wave

consider free propagation:

$$1+2 \rightarrow \psi(x,t) = \int dk A(k) e^{-i \overbrace{\frac{\hbar k^2}{2m} \cdot t}^{\omega(k)}} e^{i k \cdot x}$$

$$i \hbar \cdot \partial_t \psi(x,t) = \int dk A(k) \underbrace{\hbar \omega(k)}_{\frac{\hbar^2 k^2}{2m}} e^{-i \frac{\hbar k^2}{2m} \cdot t} e^{i k \cdot x}$$

$$= \frac{\hbar^2}{2m} \int dk A(k) e^{-i \frac{\hbar k^2}{2m} \cdot t} \underbrace{k^2 e^{i k \cdot x}}_{-\frac{\partial^2}{\partial x^2} e^{i k \cdot x}}$$

$$= - \frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \int dk A(k) e^{-i \frac{\hbar k^2}{2m} \cdot t} e^{i k \cdot x}$$

$$i \hbar \frac{\partial \psi}{\partial t} = - \frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2}$$

Constant potential ?

$$V(x) = V_0$$

classical
energy

$$E = \frac{p^2}{2m} + V_0 = \frac{\hbar^2}{2m} \cdot k^2 + V_0 \Rightarrow$$

$$\Rightarrow \hbar \omega = \frac{\hbar^2 k^2}{2m} + V_0$$

$$\Rightarrow \psi \sim e^{-i \left(\frac{\hbar^2 k^2}{2m} + V_0 \right) \frac{1}{\hbar} \cdot t} e^{i k \cdot x}$$

$$i \hbar \partial_t \psi = \left(\frac{\hbar^2 k^2}{2m} + V_0 \right) \psi = - \frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V_0 \psi$$

$\Rightarrow$ conjecture: $V_0 \rightarrow V(x)$

$$i \hbar \frac{\partial \psi}{\partial t} = - \frac{\hbar^2}{2m} \frac{\partial^2 \psi(x,t)}{\partial x^2} + V(x) \psi(x,t)$$

3D

$$i \hbar \frac{\partial \psi}{\partial t} = - \frac{\hbar^2}{2m} \left(\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} \right) + V \psi$$

$$\psi(x,t) \in \mathbb{C} !$$

$$\Delta \psi = \nabla^2 \psi = \text{div grad } \psi$$

$\uparrow$
Laplace operator

Probability

$$P(x,t) = |\psi(x,t)|^2 = \psi^*(x,t) \psi(x,t)$$

$$\int dx P(x,t) = 1$$

Q: Is probability conserved?

1D, for simplicity

$$P(x, t) = |\psi|^2 = \psi^* \psi = \psi^*(x, t) \psi(x, t)$$

$$\partial_t |\psi|^2 = \psi^* \partial_t \psi + \partial_t \psi^* \cdot \psi$$

$$\partial_t \psi = \frac{1}{i\hbar} \left(-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V \cdot \psi \right) \quad \rightarrow \quad -\frac{1}{i} \left(-\frac{\hbar^2}{2m} \frac{\partial^2 \psi^*}{\partial x^2} + V \psi^* \right)$$
$$= \frac{i\hbar}{2m} \left(\psi^* \frac{\partial^2 \psi}{\partial x^2} - \frac{\partial^2 \psi^*}{\partial x^2} \cdot \psi \right) + \frac{1}{i\hbar} \left(\psi^* V \psi - V \psi^* \cdot \psi \right)$$

$$\partial_t |\psi|^2 + \frac{\hbar}{2mi} \left(\psi^* \frac{\partial^2 \psi}{\partial x^2} - \text{c.c.} \right) = 0$$

$$\frac{\partial}{\partial x} \left(\frac{\hbar}{2mi} \left(\psi^* \frac{\partial \psi}{\partial x} - \text{c.c.} \right) \right)$$

|||
 $J(x, t)$

probability
current
density!

$\Downarrow$

$$\partial_t P(x, t) + \partial_x J(x, t) = 0$$

continuity equation

$$\frac{\partial}{\partial t} \int_{-\infty}^{\infty} dx P(x, t) = - \int_{-\infty}^{\infty} dx \partial_x J(x, t) = \left[-J(x, t) \right]_{-\infty}^{\infty} = 0$$

if $\psi(x, t) \xrightarrow{\pm\infty} 0$

$$\Rightarrow P = \int dx |\psi(x, t)|^2 = \text{const}$$

Time independent Schrödinger equation

Separation of variables

$$i \hbar \partial_t \psi(x, t) = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V(x) \psi(x, t)$$

Assume $\psi(x, t) = f(t) \cdot \varphi(x)$

$$\Rightarrow \underbrace{i \hbar \frac{\dot{f}(t)}{f(t)}}_{\parallel} = \frac{1}{\varphi(x)} \left(-\frac{\hbar^2}{2m} \frac{\partial^2 \varphi}{\partial x^2} + V(x) \varphi(x) \right)$$

$$\parallel \quad B \quad \Rightarrow \quad \dot{f} = -\frac{i}{\hbar} \cdot B \cdot f \quad \Rightarrow \quad f(t) = e^{-\frac{i}{\hbar} B \cdot t}$$

but B is energy : $B = E$

$$i \hbar \cdot \partial_t \psi = B \psi = E \cdot \psi$$

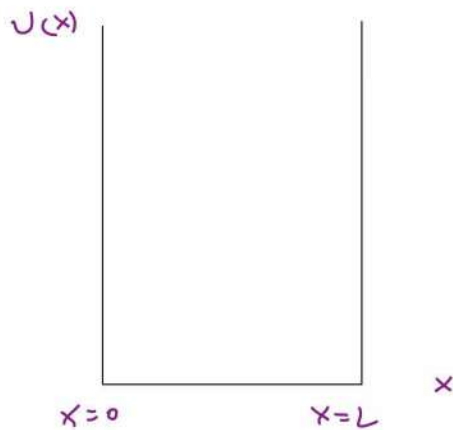
$$\psi(x, t) = e^{-\frac{i}{\hbar} E \cdot t} \varphi(x)$$

where φ satisfies

$$E \varphi(x) = -\frac{\hbar^2}{2m} \frac{\partial^2 \varphi}{\partial x^2} + V(x) \varphi(x)$$

Simple problems

A Infinite potential well



$$x \in [0, L]$$

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \varphi}{\partial x^2} = E \varphi$$

$$\Rightarrow -\frac{\partial^2 \varphi}{\partial x^2} = \kappa^2 \cdot \varphi \quad \kappa = \frac{1}{\hbar} \sqrt{2mE}$$

$$\Rightarrow \varphi(x) = \alpha \sin(\kappa \cdot x) + \beta \cos(\kappa \cdot x)$$

boundary conditions

$$a) \quad \varphi(0) = 0 \quad \Rightarrow \quad \beta = 0$$

$$b) \quad \varphi(L) = 0 \quad \Rightarrow \quad \sin(\kappa \cdot L) = 0$$

$$\Rightarrow \quad L \kappa = n \cdot \pi \quad \rightarrow \quad \kappa_n = \frac{n \cdot \pi}{L}$$

$$\Rightarrow \varphi_n(x) = A \cdot \sin\left(\frac{n\pi}{L} \cdot x\right)$$

$$\text{Normalization: } \int_0^L dx \quad A^2 \sin^2\left(\frac{n\pi}{L} \cdot x\right) = \frac{L}{2} A^2 = 1$$

$$\Rightarrow \varphi_n(x) = \sqrt{\frac{2}{L}} \sin\left(n \frac{\pi}{L} \cdot x\right)$$

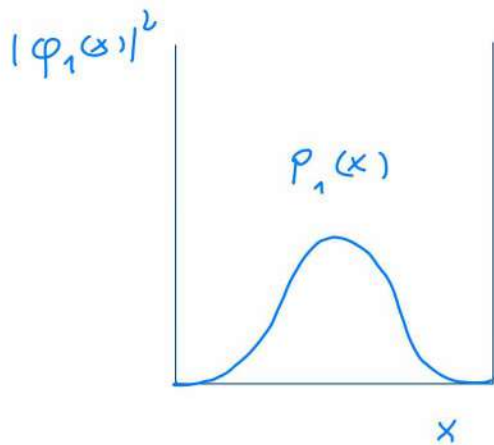
$$\Rightarrow E_n = \frac{\hbar^2}{2m} \kappa_n^2 = \frac{\hbar^2}{8 \cdot m \cdot L^2} \cdot n^2$$

$$\Rightarrow \varphi_1(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi}{L} \cdot x\right) \quad (n=1)$$

$$\psi_1(x, t) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi}{L} \cdot x\right) e^{-i \frac{E_1}{\hbar} \cdot t} \quad E_1 = \frac{\hbar^2}{8mL^2}$$

$$\rho_1(x) = |\psi_1|^2 = \frac{2}{L} \sin^2\left(\frac{\pi}{L} \cdot x\right)$$

time independent



3D infinite well

$$E \cdot \varphi(x, y, z) = -\frac{\hbar^2}{2m} \left(\frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2} + \frac{\partial^2 \varphi}{\partial z^2} \right) + V(x, y, z) \cdot \varphi$$

$$V(x, y, z) = \begin{cases} 0 & \text{if } x \in [0, L_1], y \in [0, L_2], z \in [0, L_3] \\ \infty & \text{otherwise} \end{cases}$$

solution $\varphi(x, y, z) = A \sin(k_1 x) \sin(k_2 y) \sin(k_3 z)$

$$k_1 = n_1 \frac{\pi}{L_1}, \quad k_2 = n_2 \frac{\pi}{L_2}, \quad k_3 = n_3 \frac{\pi}{L_3}$$

$$\Rightarrow \frac{\partial^2}{\partial x^2} \varphi = -k_1^2 \varphi = -n_1^2 \frac{\pi^2}{L_1^2} \cdot \varphi$$

$$\Rightarrow -\frac{\hbar^2}{2m} \Delta \varphi = \underbrace{\frac{\hbar^2}{2m} \left(n_1^2 \frac{\pi^2}{L_1^2} + n_2^2 \frac{\pi^2}{L_2^2} + n_3^2 \frac{\pi^2}{L_3^2} \right)}_{E_{n_1, n_2, n_3}} \cdot \varphi$$

$$E = \frac{\hbar^2}{2m} \left(\frac{n_1^2}{L_1^2} + \frac{n_2^2}{L_2^2} + \frac{n_3^2}{L_3^2} \right)$$